

Module 9 Week A — Stretch: SPARQL CLI

Dispatcher

Honors Track. Stretch is for learners who have completed all core assignments, are On Track or Advanced, and are attending consistently. See the Honors Track policy in TalentLMS for full eligibility.

Due: Thursday end of day. Resubmissions are accepted through Saturday of the assignment week.

Prerequisite: Base Lab 9A and Integration 9A complete. You will reuse the Fuseki stack-up from Lab 9A and the publications ontology from Integration 9A.

What you’ll build

A small command-line tool that takes a fixed-vocabulary natural-language intent and dispatches to the matching SPARQL query against the integration’s publications dataset. This is a tiny preview of the Week B integrated pipeline — a learner who builds it has done a minimum-viable NL→formal-query reduction.

Example session:

```
$ python query.py "list authors at NeurIPS"
:author000
:author017
...

$ python query.py "papers per topic"
:topic_vision-transformers 8
:topic_language-models 6
...

$ python query.py "top 5 cited"
:paper063 485
:paper043 475
...
```

Requirements

- Reuse the `data/publications.ttl` and `docker-compose.yml` from your integration repo.
- Support **at least 5 intents**, covering at least one `SELECT`, one `CONSTRUCT`, and one `ASK`.
- Use `argparse` (or `click`) — not raw `sys.argv` parsing.
- Emit a clear error message on an unknown intent, listing the supported intents.
- Document the intent → SPARQL mapping in a Markdown table in the README.
- Include a `requirements.txt` and a `pytest tests/` suite verifying that each intent’s dispatch returns the expected SPARQL structure and that an unknown intent exits non-zero with the usage banner.

Repo decision

You may either:

- Build a **standalone repo** (recommended for portfolio visibility).
- Or build on a fork of your Integration 9A repo on a feature branch.

Either way, the work must be substantively different from the integration — it is a dispatcher, not a tenth query. Reusing the eight queries from the integration as the backing SPARQL is fine and expected.

Deliverables

1. `query.py` — the dispatcher.
2. `tests/test_dispatcher.py` — your test suite.
3. `README.md` — intent → SPARQL mapping table; how to run; design notes.

Submission

1. Push your repo.
2. **Paste your repo (or PR) URL into TalentLMS → Module 9 → Stretch (Honors) — SPARQL CLI dispatcher to submit this assignment.**